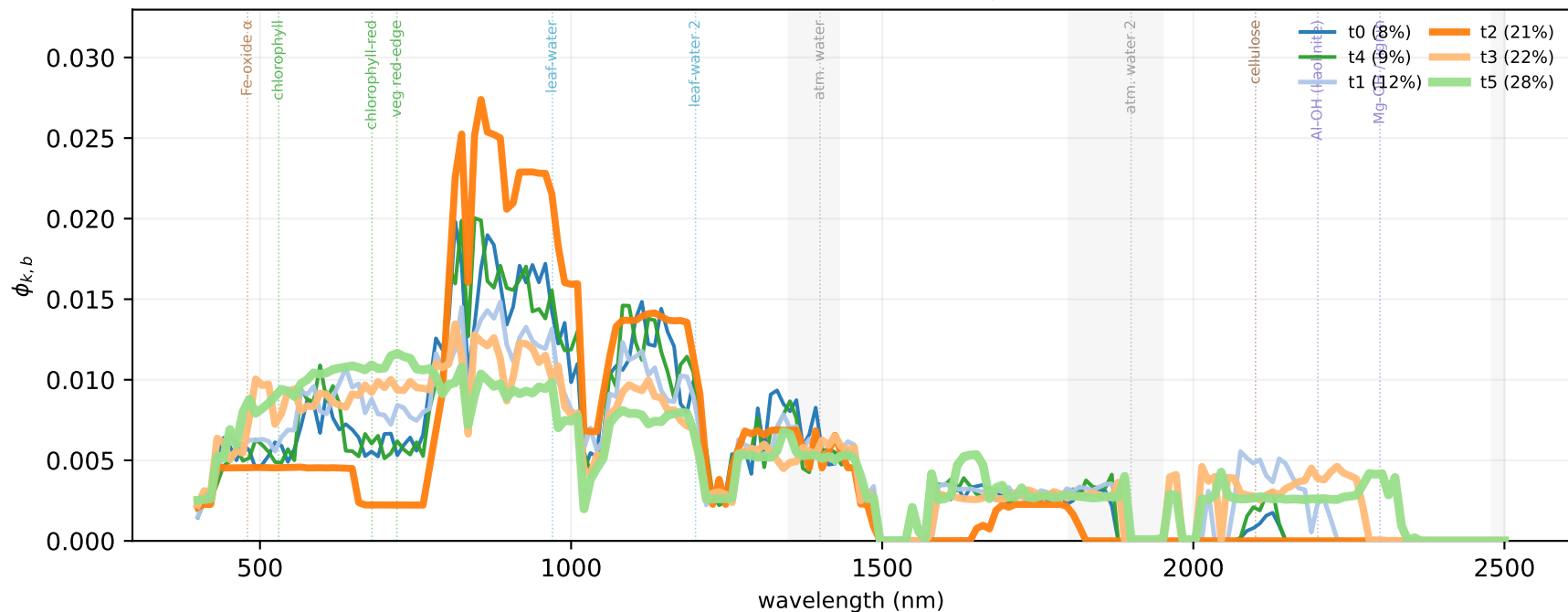


Topic spectra contrast — every ϕ_k overlaid; line width = topic prevalence; vertical lines = diagnostic absorption / reflectance features

(a) Salinas-A K=6 — band-robust (SWIR paired ARI 0.766)



(b) Kennedy Space Center K=12 — band-fragile (paired ARI ≈ 0.01 across all masks)

